

HW6_TestThat

Description

This document demonstrates how to use the Assignment 5 functions to create sample neighborhood data and calculate flood risk, social vulnerability, insurance retreat risk, and cumulative risk.

```
## Load Functions  
library(here)
```

here() starts at C:/Users/tasha/OneDrive/Desktop/UCSB (winter 25 on)/HW6_Tests

```
source(here("Create_Neighborhood_Data.R"))  
source(here("compute_flood_risk.R"))  
source(here("calculate_vulnerability.R"))  
source(here("calculate_Insurance_Retreat.R"))  
source(here("cumulative_risk_function.R"))
```

Example Neighborhood Data

```
neighborhood_data <- create_neighborhood_data(n = 10)  
  
head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	2184.6769	974.6190	142.5274	83.40960	0.2571270	0.17343338
2	b	794.7635	397.5130	137.4074	61.87006	0.3208297	0.22988344
3	c	422.7994	459.6166	246.0705	92.53511	0.4561859	0.98848231
4	d	2853.5914	814.1083	276.6307	168.36012	0.1178055	0.28372271

```

5     e  902.7243      814.7595  119.6908      130.26055 0.1262571 0.06867626
6     f 2057.1722      716.1888  168.5787      87.15340 0.7047041 0.11573993
  no_insurance  renters    elderly  children  disabled percent_nonrenewal
1  0.53575633 0.5352153 0.27403622 0.2816837 0.56216232      0.1285954
2  0.18877151 0.7054994 0.96274823 0.9588317 0.79872366      0.1668290
3  0.73467363 0.1962113 0.20039805 0.5584955 0.12796682      0.3044032
4  0.67456947 0.1846200 0.55339916 0.2520210 0.75605484      0.5909944
5  0.02483169 0.1031998 0.09422069 0.8028161 0.76037030      0.1844756
6  0.18503607 0.9422406 0.02798060 0.7988127 0.09600861      0.2796330
  premium_increase past_claims
1      0.3432284      42.55474
2      0.3718563     121.15046
3      0.4050474     195.70632
4      0.8844464     146.58512
5      0.6401678      31.50540
6      0.2332427     178.54626

```

Calculating the flood risk

```

neighborhood_data$flood_risk <- compute_flood_risk(
  elevation = neighborhood_data$elevation,
  distance_coast = neighborhood_data$distance_coast,
  mean_rain = neighborhood_data$mean_rain,
  num_past_floods = neighborhood_data$num_past_floods
)

head(neighborhood_data)

```

```

      city elevation distance_coast mean_rain num_past_floods  low_inc  minority
1      a  2184.6769      974.6190  142.5274      83.40960 0.2571270 0.17343338
2      b   794.7635      397.5130  137.4074      61.87006 0.3208297 0.22988344
3      c   422.7994      459.6166  246.0705      92.53511 0.4561859 0.98848231
4      d  2853.5914      814.1083  276.6307     168.36012 0.1178055 0.28372271
5      e   902.7243      814.7595  119.6908     130.26055 0.1262571 0.06867626
6      f 2057.1722      716.1888  168.5787      87.15340 0.7047041 0.11573993
  no_insurance  renters    elderly  children  disabled percent_nonrenewal
1  0.53575633 0.5352153 0.27403622 0.2816837 0.56216232      0.1285954
2  0.18877151 0.7054994 0.96274823 0.9588317 0.79872366      0.1668290
3  0.73467363 0.1962113 0.20039805 0.5584955 0.12796682      0.3044032
4  0.67456947 0.1846200 0.55339916 0.2520210 0.75605484      0.5909944

```

5	0.02483169	0.1031998	0.09422069	0.8028161	0.76037030	0.1844756
6	0.18503607	0.9422406	0.02798060	0.7988127	0.09600861	0.2796330
	premium_increase	past_claims	flood_risk			
1	0.3432284	42.55474	0.2966292			
2	0.3718563	121.15046	0.4695990			
3	0.4050474	195.70632	0.5971389			
4	0.8844464	146.58512	0.5817885			
5	0.6401678	31.50540	0.4657963			
6	0.2332427	178.54626	0.4032636			

Calculating the Social Vulnerability

```
neighborhood_data$vulnerability_score <- calculate_vulnerability(
  low_inc = neighborhood_data$low_inc,
  minority = neighborhood_data$minority,
  no_insurance = neighborhood_data$no_insurance,
  renters = neighborhood_data$renters,
  elderly = neighborhood_data$elderly,
  children = neighborhood_data$children,
  disabled = neighborhood_data$disabled
)

head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	2184.6769	974.6190	142.5274	83.40960	0.2571270	0.17343338
2	b	794.7635	397.5130	137.4074	61.87006	0.3208297	0.22988344
3	c	422.7994	459.6166	246.0705	92.53511	0.4561859	0.98848231
4	d	2853.5914	814.1083	276.6307	168.36012	0.1178055	0.28372271
5	e	902.7243	814.7595	119.6908	130.26055	0.1262571	0.06867626
6	f	2057.1722	716.1888	168.5787	87.15340	0.7047041	0.11573993
	no_insurance	renters	elderly	children	disabled	percent_nonrenewal	
1	0.53575633	0.5352153	0.27403622	0.2816837	0.56216232	0.1285954	
2	0.18877151	0.7054994	0.96274823	0.9588317	0.79872366	0.1668290	
3	0.73467363	0.1962113	0.20039805	0.5584955	0.12796682	0.3044032	
4	0.67456947	0.1846200	0.55339916	0.2520210	0.75605484	0.5909944	
5	0.02483169	0.1031998	0.09422069	0.8028161	0.76037030	0.1844756	
6	0.18503607	0.9422406	0.02798060	0.7988127	0.09600861	0.2796330	
	premium_increase	past_claims	flood_risk	vulnerability_score			
1	0.3432284	42.55474	0.2966292	0.3742020			

2	0.3718563	121.15046	0.4695990	0.5950411
3	0.4050474	195.70632	0.5971389	0.4660591
4	0.8844464	146.58512	0.5817885	0.4031704
5	0.6401678	31.50540	0.4657963	0.2829103
6	0.2332427	178.54626	0.4032636	0.4100747

Calculating Insurance Retreat Score

```
neighborhood_data <- calculate_insurance_retreat_score(neighborhood_data)

head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	2184.6769	974.6190	142.5274	83.40960	0.2571270	0.17343338
2	b	794.7635	397.5130	137.4074	61.87006	0.3208297	0.22988344
3	c	422.7994	459.6166	246.0705	92.53511	0.4561859	0.98848231
4	d	2853.5914	814.1083	276.6307	168.36012	0.1178055	0.28372271
5	e	902.7243	814.7595	119.6908	130.26055	0.1262571	0.06867626
6	f	2057.1722	716.1888	168.5787	87.15340	0.7047041	0.11573993

	no_insurance	renters	elderly	children	disabled	percent_nonrenewal
1	0.53575633	0.5352153	0.27403622	0.2816837	0.56216232	0.1285954
2	0.18877151	0.7054994	0.96274823	0.9588317	0.79872366	0.1668290
3	0.73467363	0.1962113	0.20039805	0.5584955	0.12796682	0.3044032
4	0.67456947	0.1846200	0.55339916	0.2520210	0.75605484	0.5909944
5	0.02483169	0.1031998	0.09422069	0.8028161	0.76037030	0.1844756
6	0.18503607	0.9422406	0.02798060	0.7988127	0.09600861	0.2796330

	premium_increase	past_claims	flood_risk	vulnerability_score
1	0.3432284	42.55474	0.2966292	0.3742020
2	0.3718563	121.15046	0.4695990	0.5950411
3	0.4050474	195.70632	0.5971389	0.4660591
4	0.8844464	146.58512	0.5817885	0.4031704
5	0.6401678	31.50540	0.4657963	0.2829103
6	0.2332427	178.54626	0.4032636	0.4100747

	insurance_retreat_score
1	0.2312842
2	0.3366246
3	0.4794866
4	0.7367614
5	0.3613627
6	0.3836966

Calculate Cumulative Risk

```
neighborhood_data$cumulative_risk_index <- compute_risk_index(
  flood_vulnerability = neighborhood_data$flood_risk,
  social_vulnerability = neighborhood_data$vulnerability_score,
  non_coverage_risk = neighborhood_data$insurance_retreat_score
)

head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	2184.6769	974.6190	142.5274	83.40960	0.2571270	0.17343338
2	b	794.7635	397.5130	137.4074	61.87006	0.3208297	0.22988344
3	c	422.7994	459.6166	246.0705	92.53511	0.4561859	0.98848231
4	d	2853.5914	814.1083	276.6307	168.36012	0.1178055	0.28372271
5	e	902.7243	814.7595	119.6908	130.26055	0.1262571	0.06867626
6	f	2057.1722	716.1888	168.5787	87.15340	0.7047041	0.11573993

	no_insurance	renters	elderly	children	disabled	percent_nonrenewal
1	0.53575633	0.5352153	0.27403622	0.2816837	0.56216232	0.1285954
2	0.18877151	0.7054994	0.96274823	0.9588317	0.79872366	0.1668290
3	0.73467363	0.1962113	0.20039805	0.5584955	0.12796682	0.3044032
4	0.67456947	0.1846200	0.55339916	0.2520210	0.75605484	0.5909944
5	0.02483169	0.1031998	0.09422069	0.8028161	0.76037030	0.1844756
6	0.18503607	0.9422406	0.02798060	0.7988127	0.09600861	0.2796330

	premium_increase	past_claims	flood_risk	vulnerability_score
1	0.3432284	42.55474	0.2966292	0.3742020
2	0.3718563	121.15046	0.4695990	0.5950411
3	0.4050474	195.70632	0.5971389	0.4660591
4	0.8844464	146.58512	0.5817885	0.4031704
5	0.6401678	31.50540	0.4657963	0.2829103
6	0.2332427	178.54626	0.4032636	0.4100747

	insurance_retreat_score	cumulative_risk_index
1	0.2312842	0.3115960
2	0.3366246	0.4892506
3	0.4794866	0.5338369
4	0.7367614	0.5480779
5	0.3613627	0.3874287
6	0.3836966	0.4022728

The Final Output

```
final_output <- neighborhood_data[
  c("city", "flood_risk", "vulnerability_score",
    "insurance_retreat_score", "cumulative_risk_index")
]

final_output
```

	city	flood_risk	vulnerability_score	insurance_retreat_score
1	a	0.2966292	0.3742020	0.2312842
2	b	0.4695990	0.5950411	0.3366246
3	c	0.5971389	0.4660591	0.4794866
4	d	0.5817885	0.4031704	0.7367614
5	e	0.4657963	0.2829103	0.3613627
6	f	0.4032636	0.4100747	0.3836966
7	g	0.3831709	0.5338201	0.3027507
8	h	0.2824987	0.3680055	0.4842128
9	i	0.5537072	0.4146516	0.3447423
10	j	0.7322809	0.4284787	0.8673445

	cumulative_risk_index
1	0.3115960
2	0.4892506
3	0.5338369
4	0.5480779
5	0.3874287
6	0.4022728
7	0.4199839
8	0.3446200
9	0.4725278
10	0.6535241

Test the functions

```
library(testthat)
```

Warning: package 'testthat' was built under R version 4.4.3

```
# run all the test files
test_dir("tests")
```

```
v | F W S OK | Context
```

```
/ |          0 | calculate_insurance_retreat_score
v |          3 | calculate_insurance_retreat_score
```

```
/ |          0 | calculate_vulnerability
v |          2 | calculate_vulnerability
```

```
/ |          0 | compute_flood_risk
v |          3 | compute_flood_risk
```

```
/ |          0 | create_neighborhood_data
v |          2 | create_neighborhood_data
```

```
/ |          0 | cumulative_risk_func
v |          2 | cumulative_risk_func
```

```
== Results =====
[ FAIL 0 | WARN 0 | SKIP 0 | PASS 12 ]
```

Explanation

This example shows how all five of the Assignment 5 modules can be used together to assess flood/social vulnerability at the neighborhood level. First, sample neighborhood data is generated using `create_neighborhood_data()`. Then, flood risk, social vulnerability, and insurance retreat score are calculated separately, but require inputs from the following modules to run accordingly. Finally, each metric is combined into a cumulative risk index through the `compute_risk_index()` function. This provides a measure of the overall neighborhood risk!